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# A Fast 3D Modeling of Objects with Mobile Devices

## 1. Background of the Research Project

The content of this part is the explanation of the importance of this research and its uniqueness and advantages to the current studies. The contents also consist of ideas from other sources to create an original and unique approach to this research.

### 1a. Importance

Manual 3D object modeling costs a significant amount of time using modeling software. There have been special sensors for 3D imaging like Lidar and Kinect [1] in order to generate objects and the surrounding just by scanning. Recently, time-of-flight (TOF) [1,2] cameras have been made which is a good alternative for the sensors mentioned and are starting to be embedded on mobile devices. With the help of TOF cameras, it is possible to build a 3D object modeling system by turning a mobile device into a scanner. Thereafter, the proposed project can provide a fast 3D modeling of objects using accessible devices.

### 1b. Uniqueness and Advantages

The advantage of our model is that it is accessible because most of the population held a mobile device. The resulting 3D model of objects scanned and generated through a mobile device and cloud computing does not remain in the device but is accessible and shareable to other devices and high-end computers. There have been studies related to 3D modeling without mobile devices like ARCore [3] of Google and 3D Modeling Kit [4] of Huawei, but each has its disadvantages. ARCore can generate raw depth and confidence value information to generate a depth map for the RGB image but this idea is complex enough for our model. ARCore focuses on augmented reality and relies on motion tracking in attaining depth information thus the model does not fit into ours. On the other hand, the modeling kit focuses on the images without depth thus requiring detailed textures on images to be used. The kit is 3D reconstruction capable but complex enough to consider no-depth images. The proposed model aims to achieve a fast modeling system utilizing mobile phone sensors and cloud computing to produce a high-quality 3D model of a scanned object.

# 2. Review of Related Literature

The purpose of this review is to generate a new method for 3D reconstruction designed for mobile devices. The information will also support the complexity of the new method proposed after it is made. The contents below are related to 3D modeling, processing, and reconstruction.

In computer graphics, 3D modeling is the process of creating a 3D representation of objects and surfaces. The model can be made using a collection of vertices (points) joined together in a 3D grid using polygonal shapes forming a mesh [5]. Alternatively, points are enough to visualize an object or a surface calling the collection a point cloud. In a grid system, a point can be called a voxel the same as the pixel but in 3D representation [6]. A point cloud is a collection of points with XYZ coordinate system values [7]. Each point comes from the scanned image with depth information. Collecting and estimating these points to generate a 3D model is called 3D reconstruction [8].

Images used as sources for the 3D model reconstruction can simply use any type of image even in grayscale but best when the image has depth information. Images other than images with depth information still end up calculating the depth of the elements found in the image before converting it into a 3D model like the use of monocular cues or stereo cameras. Thus, the study utilizes the time-of-flight (TOF) camera sensor available on mobile devices. TOF cameras provide 3D imaging that uses a modulated light source and observes the reflected light [9]. The phase shift between the source and the reflection is then translated into a depth map. With the TOF camera embedded on mobile devices, the devices can be used as a scanner to produce RGB-D images for 3D reconstruction.

Reconstructing a 3D model of a point cloud out of one image is simpler compared to real-time scanning. The latter, real-time scanning, can reconstruct models with greater information than the former by fusing all the images captured. The fusion of images is complex leading to another set of studies. Three examples of these algorithms are BundleFusion [10], KinectFusion [11], and Voxel Hashing [12] which reconstructs 3D models using depth cameras. Furthermore, libraries that have the capabilities of real-time reconstruction are widely available. Open3D [13] and PCL [14] are well-known libraries for point cloud 3D reconstruction.

Fusing the images into a 3D model uses an algorithm that aligns two (2) point clouds is called registration. Camera or other imaging sensors have limited view range thus registration algorithm is required to generate a 3D representation of an object or surrounding that can not be captured by a single 2D frame [8]. Point cloud registration outputs the best estimate of the transformation matrix of the source or partial scan point cloud to the reference point cloud and this process repeats until the scan is complete producing a complete 3D model. There must be a set of corresponding points, called correspondence, from source to the reference in order to calculate the transformation. Furthermore, the transformation estimation and correspondence search have unique characteristics. The best transformation matrix can be calculated if the correspondence point set is available, meanwhile, the correspondence set can be searched given a transformation matrix. This leads to the whole process being conducted alternatively until the geometric projection error is minimum [15].

Models made in point clouds also have their disadvantages. One of which is the file size. This can be a problem to the study later that needs to be addressed because the proposed model's output processed on the cloud is to be delivered back to the client's side. There are algorithms that try minimizing the said problem calling the process downsampling. Technically, downsampling takes a certain percentage of points in the original model without destroying the model's shape. The simplest algorithm made is through the random selection of a set of points. On the other hand, the best ones use a grid method that lessens the point through each grid or box. The first algorithm might not be efficient for this study because the resulting sample model through random means ends with a non-consistent model or can not generate the same model twice, thus, the latter best fits. There are two popular algorithms that use the grid method calling the two uniform and non-uniform. The former uses a grid average amongst the points while the latter tries to retain a certain number of points in each grid. The former is best only in retaining the shape of the model while the latter both retains the shape of the model and density of points but needs more time than the former. The study only needs the model's shape to create a mesh model, therefore, the study will use the uniform grid method in downsampling.

A mesh and a point cloud can both represent an object or scene in a 3D model. The difference between the two is that a mesh uses points as vertices to form polygons while a point cloud is only a collection of points. As a collection of vertices and polygons, a mesh is a 3D representation from unfinished shapes, clustered points, or even abstract forms of vertices to create a single form or mesh [16]. In this paper, the structured mesh will be created from clustered data of point clouds to create an object from the scanned images of a mobile device. Furthermore, cleaning the point cloud data is one of the key problems in creating a mesh, one method to simplify the mesh creations is using an error metric based on Hausdorff Distance in order to keep and extract the featured points of the point cloud [17]. The points are classified based on their geometric distribution of point neighborhoods. The data classification of points determines the quality and the efficiency of subsequent reconstruction in relation to the importance of mesh reconstruction [17].

# 3. Originality

A fast 3D modeling system of objects with the use of mobile devices is the goal of the study. Its scope starts from object scanning with the mobile device to its reconstruction in the cloud until the 3D model is ready to be delivered back to the device. Technically, an RGB-D image is needed for the 3D model to be reconstructed, therefore, the mobile devices that act as scanners must have a range imaging camera like the time-of-flight (TOF) camera available for mobile devices. On the other hand and the most important part, the 3D reconstruction needs a lot of computing power that mobile devices cannot do within seconds or minutes. Cloud computing is needed for the processing part of the system for faster modeling. Furthermore, the model that will be reconstructed is in the form of a point cloud thus delivery of the model back to the device might be delayed more due to the file size thus conversion to a mesh is done as a part of the whole system's process. When the  3D model is returned to the mobile device, it is now ready to be used or shared to other devices or computers.

# 4. Materials and Methods

Our approach uses a range imaging capable mobile devices to generate RGB-D images to be converted into a 3D model. Smartphones are the main interface of the project both from input and output. On the other hand, for the processing, mobile devices have limited processing power, therefore, cloud computing will be utilized in order to generate models faster. Python programming language will be used for faster modeling of the proposal.

The model is composed of 3 major processes in 3D model creation based on RGBD data. The first part is raw data processing. A mask is made and used to filter out data other than the object in the depth map. Using the filtered depth maps and the RGB images, a 3D point cloud model is made in the second process. The integration and de-integration technique of BundleFusion [18] is needed to create a dynamic and less error reconstruction. The technique adds more confidence to the resulting model as redundancy of object information leads to less error and high-quality reconstruction. Lastly, in the third process, a mesh is created to save memory space. The point classification method on the Geometric Distribution of Point Neighborhood [17] is used to create a mesh from point cloud data and a simplification using Hausdorff Distance [17] is used to simplify the geometric features of the point cloud. Moreover, downsampling is done first to the current point cloud model to minimize memory consumption.

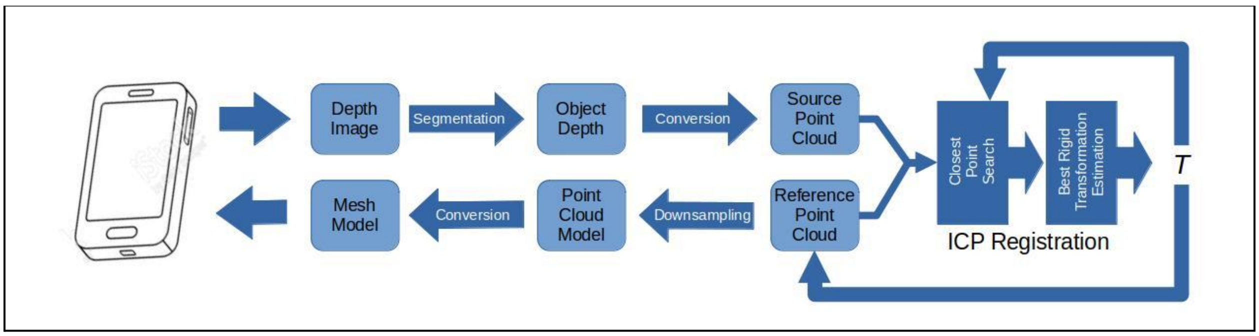



Figure 1: Architectural design of fast 3D modeling of objects. An object is scanned with a mobile device. The images from a scan are then processed to generate a 3D point cloud model that is to be converted into a mesh model.

## 4a. Object Masking

Depth maps can easily be segmented through intensity thresholding. Using a histogram, the depth map intensities can be visually represented in multimodal distribution. The first distribution means the nearest and the intensity range that represents the object to be filtered. To get the said range, the valley between the first and second distributions must be determined. In which, the valley is the threshold that separates the two distributions represented as $T_0$. The threshold can be calculated using the adaptive thresholding algorithm [19] which calculates the threshold dynamically on different images. A mask can be generated using the calculated threshold value as

$$M = \{0 \text{ if } D>T_0 \text{ else } 1\},$$

where M is the mask representing the subject region and D is the depth map values.

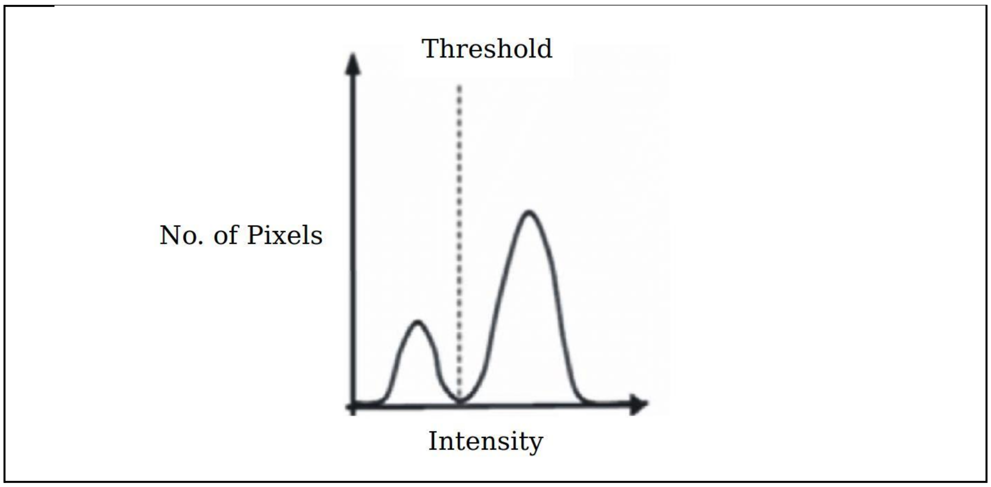



Figure 2: Object segmentation from the depth image. The first distribution represents the object while the others represent the background.

## 4b. Integration and De-Integration

Integration and de-integration of depth frames fix errors in the volumetric representation as soon as better estimates are available. Given a set of point or voxel distances D and weight W, we can integrate the frame by

$$D' = (DW + w_i d_i) / (W + w_i)\ ,\ W' = W + w_i.$$

Reversing this operation to de-integrate a frame can be done through

$$D' = (DW - w_i d_i) / (W + w_i)\ ,\ W' = W - w_i.$$

With these formulae, reconstruction in high quality can be obtained due to continuous change or revisiting.

## 4c. Mesh Creation

Converting a point cloud data to a mesh enhances the 3D object capture of the scanned objects. A voxel structure is introduced to extract spatial features from a given point cloud. A voxel network can efficiently represent a relatively fast spatial structure with its inference speed [20].

The conversion is done by basing the geometric distribution of the point neighborhood to reconstruct the point cloud to mesh, where the point cloud is divided into a boundary point and interior point where the Euclidean distance between the two points is:

$$D = \sqrt{(x_i - x)^2 + (y_i - y)^2 + (z_i - z)^2}$$

After the point cloud is divided, the flat point set, selects the seed triangle $S_0$, the 3 edges will be added to $S_0$ to the active edge queue, and sets the T value from the vertices of the triangle to one, then determine if the active edge is expandable, if not delete the queue head element and test the next element until it is expandable or it terminates itself if it is empty.

The expansion point is searched to determine if the edge is a border edge or not, if it is the queue will be deleted if the active queue is not empty, then determine it again if the active edge is expendable, if not then enter a high curvature point set for mesh reconstruction. The list of the inner edge adjacency points is updated for the two endpoints of the current edge.

Since the target is only the object itself, the expected file size would be determined by the size of the object that is scanned through the phone. The point cloud data can be simplified based on Hausdorff Distance where:

$$D(A,B)=max(d(A,B),d(B,A))$$

Which directly affects the time cost and result of the simplification. The reconstruction time will also be reduced and the data quantity will also decrease while the reconstruction efficiency gradually improves.

# 5. Expected Results

## 5a. The project plans to accomplish the following:

A. Convert the mobile device as an object scanner that can give inputs to the 3D reconstructor in RGB-D image form.

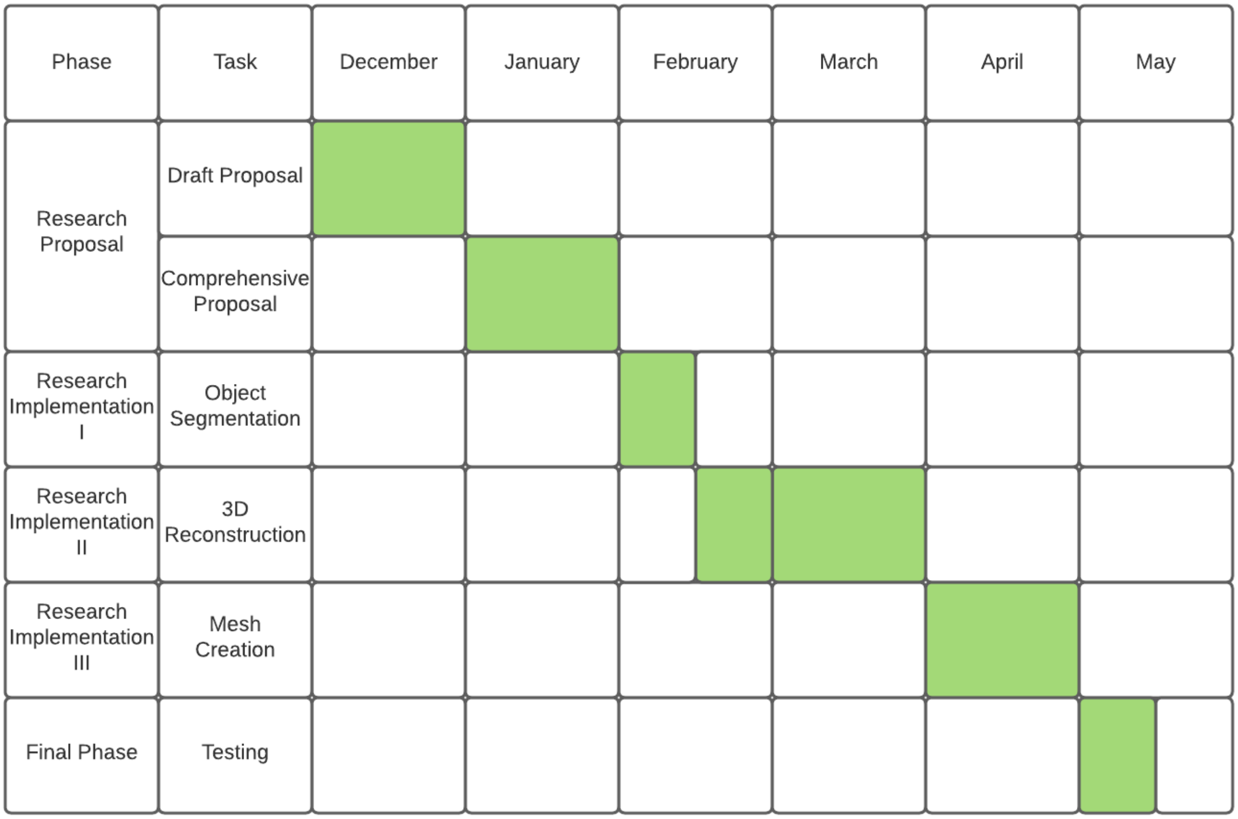
B. Filter the depth information of the object to prevent reconstructions of other elements other than the object present in the images.

C. Fuse all the images or frames to generate a high-quality 3D point cloud model.

D. Convert the point cloud model into a mesh model to minimize the delivery cost of the model back to the mobile device.

E. Test the effectiveness and efficiency of the system through comparison with existing modeling systems both in quality and speed.

# 6. Gantt Chart

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